

Learning to Refer: How Scene Complexity Affects Emergent Communication in Neural Agents

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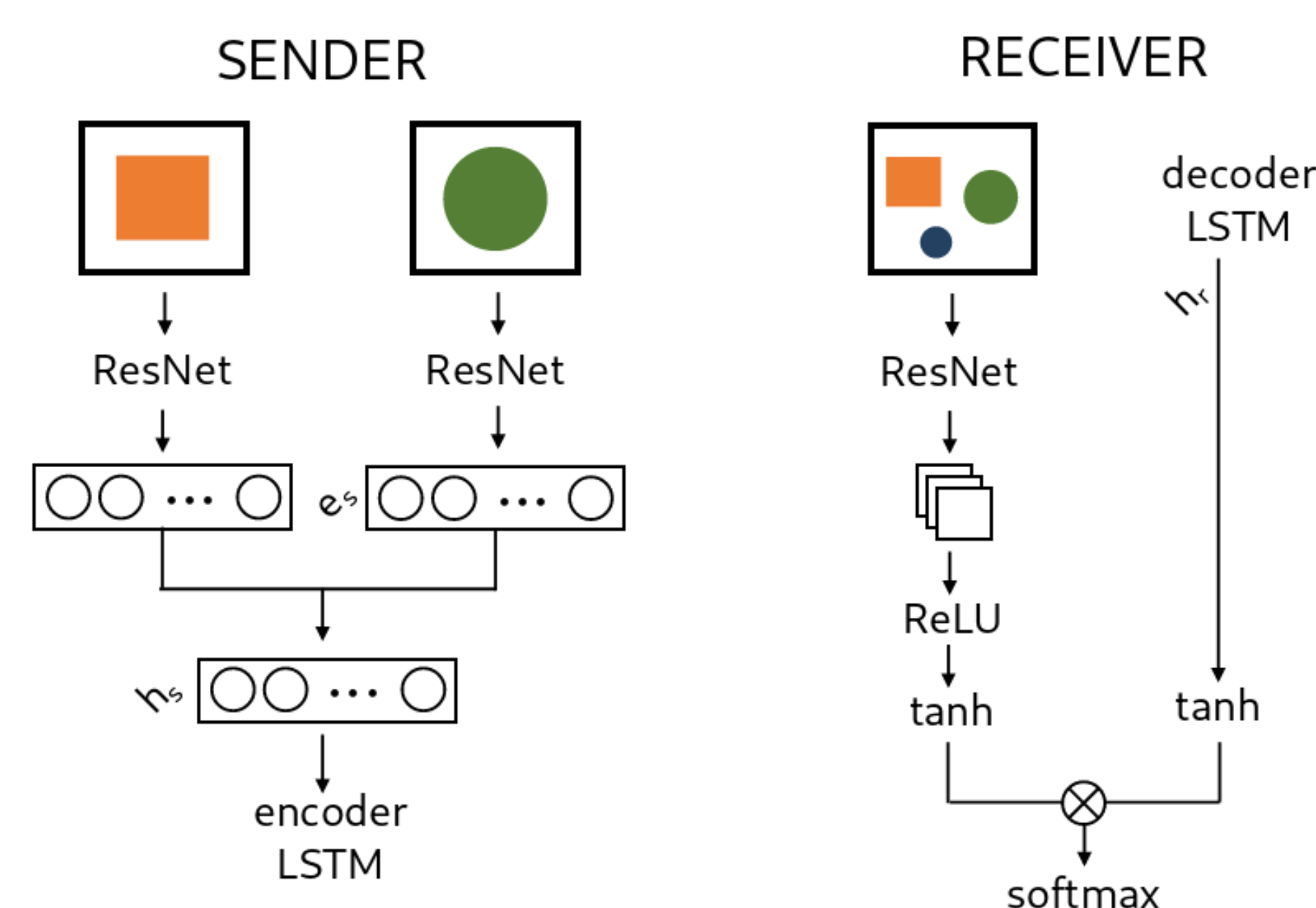
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AIMS

A picture can say thousand words...

- Study **emergence of discrete linguistic symbols** and **languages** from continuous sensory input in **interactive communication**
- Study **constraints** on symbol grounding in interacting neural agents
- Investigate the **role of scene complexity, discriminating features present, and message space** on emergent languages

SETUP



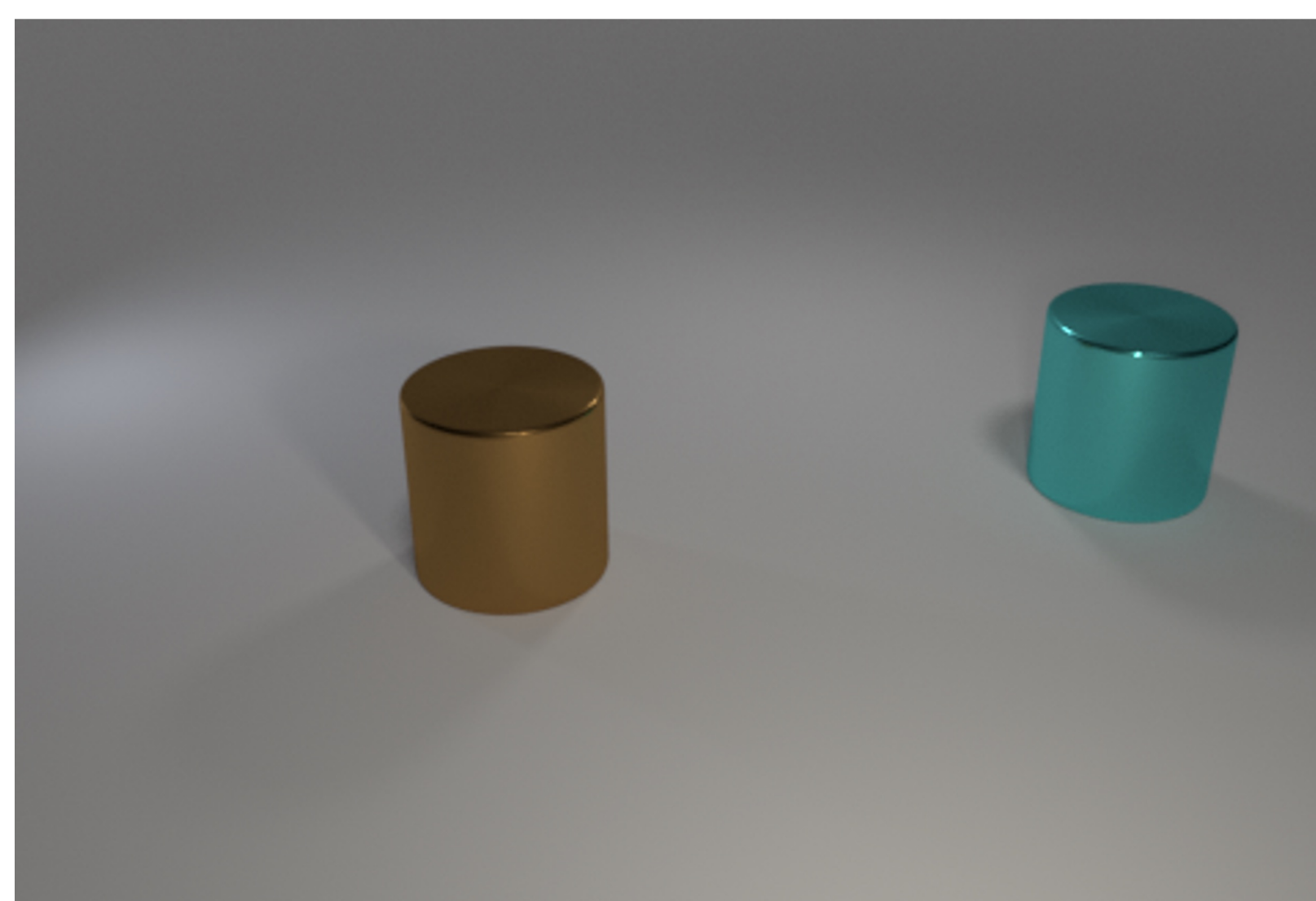
- Receiver **attends to region** in visual scene **based on the sender's message**

CONCLUSION

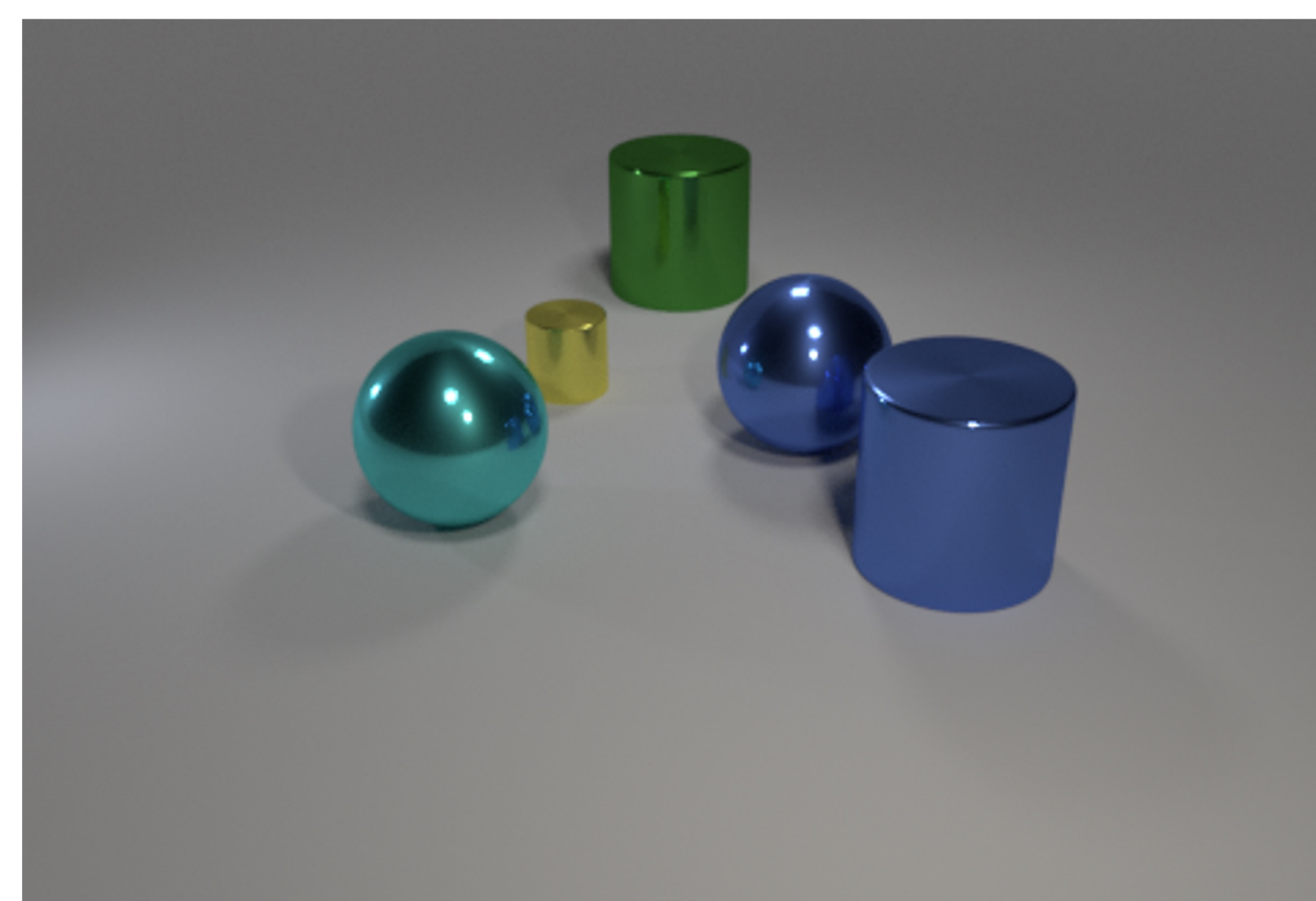
- **Medium vocabularies** ($|V| \in \{10, 16\}$) and message lengths ($n \in \{2, 4\}$) perform best
- Scene complexity (**number of distractors, number of features**) dramatically impacts communication success
- **Shape** attributes **easiest** to learn, **size** attributes **most challenging**
- Future: Analyze linguistic properties of emergent languages

EXPERIMENTS

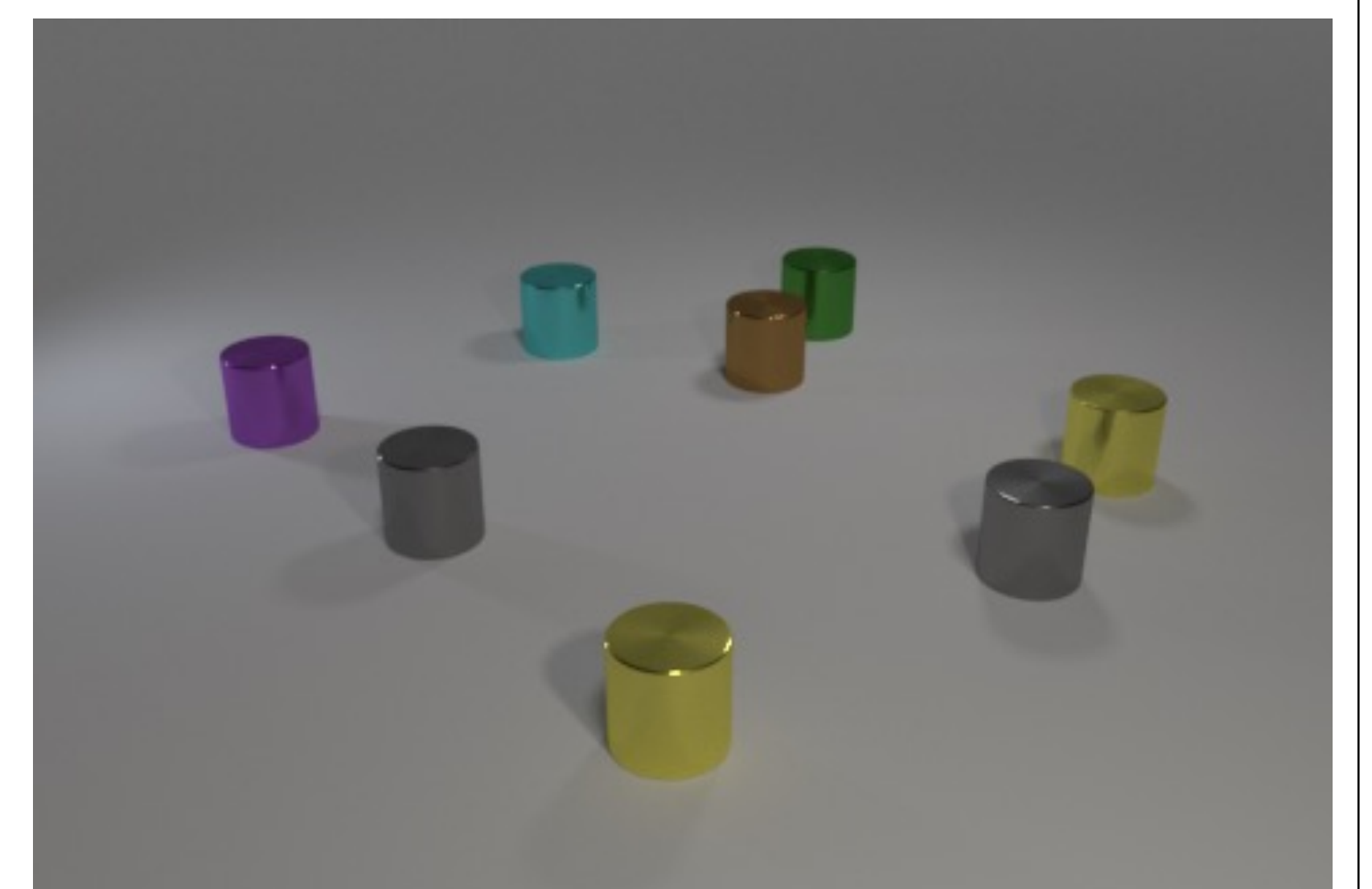
Dale-2



Dale-5



CLEVR Color



SENDER IN SUPERVISED GENERATION

Strict string accuracy	99%	69%	93%
F1 score	98,57%	89,53%	95,17%
Shape Precision	99,75%	98,30%	100%
Color Precision	98,09%	92,44%	92,76%
Size Precision	98,73%	69,43%	N/A (not used)

SENDER AND RECEIVER IN INTERACTION

